

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain Carnot cycle with P-V and T-S diagram. (CO2)
- Q.24 Explain the construction and working of Centrifugal Pump with the help of neat sketch. (CO4)
- Q.25 Discuss the construction & working Reciprocating Air Compressor. (CO4)

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3rd Sem / Automobile

Subject : Basics of Thermodynamics, Hydraulics and Pneumatics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Boyle's law is applied when _____ is constant. (CO1)
- a) Temperature b) Pressure
- c) Volume d) None of above
- Q.2 Enthalpy is the heat supplied to a system at (CO2)
- a) Constant pressure
- b) Constant temperature
- c) Constant entropy
- d) Constant volume
- Q.3 The envelop which separates thermodynamics system from surroundings is called (CO1)
- a) Boundary b) Universe
- c) State d) Process

- Q.4 Bernoulli's theorem deals with the law of conservation of (CO1)
 a) Mass b) Energy
 c) Momentum d) None of above
- Q.5 Hydraulic Ram works on the principle of (CO2)
 a) Water hammer b) Centrifugal action
 c) Reciprocating action d) All of the above
- Q.6 If the fluid particles move in a straight line the flow is (CO2)
 a) Laminar flow b) Turbulent flow
 c) Steady flow d) Uniform flow

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Write the S.I unit of intensity of pressure of liquid? (CO1)
- Q.8 A device which is used to increase the kinetic energy at the expense of pressure energy is called _____ (CO1)
- Q.9 The efficiency of Otto cycle is _____ (more/less /equal) than diesel cycle under similar condition. (CO2)

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- Q.10 Write the principle on which Hydraulic press works. (CO2)
- Q.11 $H=U+$ _____ (CO2)
- Q.12 Name the device used to measure the pressure of liquids.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain closed and isolated systems. (CO3)
- Q.14 Explain first law of thermodynamics. (CO2)
- Q.15 Explain Specific volume and Specific gravity. (CO2)
- Q.16 Explain the relationship between Absolute pressure, Atmospheric pressure and Gauge pressure. (CO3)
- Q.17 Explain the U-tube manometer with the help of neat sketch. (CO4)
- Q.18 Explain rate of flow and its unit. (CO2)
- Q.19 Explain Bernoulli's theorem. (CO3)
- Q.20 Differentiate between centrifugal pump and reciprocating pump. (CO2)
- Q.21 Write down the advantages and disadvantages of pneumatic system. (CO2)
- Q.22 Differentiate between pneumatic screw driver and pneumatic wrenches. (CO2)

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